Jungseob **Lee**

NLP Researcher · Ph.D. Student, Korea University

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Summary

I do my research in a university lab, with far fewer GPUs than the companies that build the largest models. Working this way, I have seen how much a good answer from AI still depends on computing cost. I think a grandmother in a small town in Korea who asks about a symptom late at night should get the same good answer as an engineer in Silicon Valley, and get it just as fast. So I work on LLM efficiency, making models cheaper to train and faster to run, without making them worse.

Research Interests: Large Language Models, Agentic LLM, LLM Efficiency, Efficient Reasoning

Education

Korea University *Seoul, Korea*
Integrated M.S./Ph.D. in Computer Science *2021 – Present*
NLP Lab, Advisor: Prof. Heuiseok Lim · **Expected: Feb. 2027**

Experience

VAETKI, Korean Sovereign AI Foundation Model *Jun 2025 – Jan 2026*
Post-training Project Manager, Korea University Representative
Led post-training (SFT, RL, dataset curation) for Korea University in the NC AI (NCSOFT)-led 14-institution consortium of the Ministry of Science and ICT's national program. Open-sourced in Dec 2025, state of the art on CLIcK, KoBALT, KMMLU-Pro, and IFEval. [Model] [Technical Report]

Multilingual Post-Training Datasets, Sovereign AI Foundation Model Program *Aug 2026*
Dataset Lead, Korea University with NC AI
Led the construction and release of two post-training datasets on AI Hub, Korea's national AI data platform. The High-Quality Multi-Turn Dataset holds multi-turn dialogues in Korean, English, Japanese, and Chinese across six domains (664M tokens), and the Synthetic & Instruction Dataset holds 858,960 model-synthesized instruction-response samples with chain-of-thought reasoning (5.8 GB). Download is open to Korean nationals only.
[Multi-Turn Dataset] [Synthetic & Instruction Dataset]

KURE-v3, Korean-English Retrieval Embedding Model *2026*
Core Contributor, Data Synthesis, Training Optimization, and Experiments
Synthesized the training data (3M+ query-document triplets with mined hard negatives), optimized training (teacher-score distillation, Matryoshka embeddings), and ran the experiments for the 1.1B dense retriever of the Korea University NLP & AI Lab, released in Oct 2026 with open weights and an API. It scores 0.803 nDCG@10 on MTEB(kor, v2) Retrieval, above much larger retrievers such as Qwen3-Embedding-8B. [Model] [GitHub]

Polyglot, Open-Source Multilingual Language Models *2022*
Open-Source Contributor, EleutherAI
Worked on efficient language transfer of pretrained language models with cross-lingual post-training (XPT) and MAD-X adapters, and also took part in pretraining. [GitHub]

Awards & Honors

Outstanding Reviewer Award *2026*
The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego

Best Paper Award (Top 1) *2025*
Annual Conference on Human and Cognitive Language Technology (HCLT 2025)
For *KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification* (first author)

Minister of Science and ICT Award *2023*
ICT Challenge 2023, Ministry of Science and ICT (organized by IITP)
For *KoCheckGPT*, Korean-specialized detection of LLM-generated documents, built with M. Kang and S. Lee and selected from 79 teams across 28 universities

Selected Publications

Underlined = me, * equal contribution. Full list (43 papers) on Google Scholar.

Top-Tier Conferences

The Hallucination Signal Is a Mean Shift: Why Simple Probes Suffice
J. Lee, J. Seo, H. Lim [arXiv] *EMNLP 2026 (Main)*

Language Chain in Alignment: Cross-Lingual Ranking Preference Optimization
S. Lee, M. Kim, **J. Lee**, H. Lim [arXiv] [Code] *EMNLP 2026 (Main)*

DART: Draft-Agreement Routing for Training-Free Adaptive Thinking Budgets in Hybrid Reasoning Models
J. Lee, S. Hong, S. Lee, *et al.* [arXiv] [Code] *Findings of EMNLP 2026*

ThinkFuse: Trajectory-Aware Test-Time Fusion for Small Reasoning Models
M. Kang*, **J. Lee***, J. Seo, H. Lim [arXiv] [Code] *Findings of EMNLP 2026*

Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim [PDF] *ICLR 2026 (Poster)*

Cross-lingual Optimization for Language Transfer in Large Language Models
J. Lee, S. Hong, H. Moon, H.S. Lim [PDF] *ACL 2025 (Oral)*

Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation
J. Lee, H. Moon, S. Lee, *et al.* [PDF] *Findings of ACL 2024*

PEEP-Talk: A Situational Dialogue-based Chatbot for English Education
S. Lee, Y. Jang, C. Park, **J. Lee**, *et al.* [PDF] *ACL 2023 (Demo)*

International Journals

Towards Enhancing Natural Language Inference for Identifying Erroneous Outputs in Data-to-Text Generation
C. Ok, E.K. Lee, H. Ju, **J. Lee**, D. Oh [DOI] *Knowledge-Based Systems, 2026*

An Analysis on Language Transfer of Pre-trained Language Model with Cross-lingual Post-training
S. Son*, C. Park*, **J. Lee***, *et al.* [DOI] *Expert Systems with Applications, 2025*

Analysis of the Effectiveness of Model, Data, and User-Centric Approaches for Chat Application: A Case Study of BlenderBot 2.0
C. Park*, **J. Lee***, S. Son, *et al.* [DOI] *Applied Sciences, 2024*

A Survey on Evaluation Metrics for Machine Translation
S. Lee*, **J. Lee***, H. Moon, *et al.* [PDF] *Mathematics, 2023 (142 citations)*

Other International Conferences

Skin-Deep: A Geometric Diagnostic for Alignment Fragility in Large Language Model Representations
D.J. Lee*, **J. Lee***, S. Lee, *et al.* [arXiv] [Code] *Findings of AACL-IJCNLP 2026*

Under Review

Predicting and Repairing Merge Collapse in Large Language Models
J. Lee, S. Lee, S. Eo, *et al.* [arXiv] [Code] *arXiv, Oct 2026*

Distilling Directional Verification
J. Lee, S. Eo, S. Hong, *et al.* [arXiv] [Code] *arXiv, Oct 2026*

CAST: Cost-Aware Speculative Trees from One-Pass Block Drafters
J. Lee, S. Eo [arXiv] [Code] *arXiv, Oct 2026*

Refusal Localizes, the Damage Relocates: Safety Layers Under Few-Sample Fine-Tuning
J. Lee*, D.J. Lee*, S. Eo, *et al.* [arXiv] [Code] *arXiv, Oct 2026*

Recovering Off-Policy Supervision for Speculative Decoding
J. Lee, C. Park, S. Eo, H. Moon [arXiv] [Code] *arXiv, Sep 2026*

Risk-Controlled Selective LLM Answering by Pricing Label-Free Checks
D.J. Lee*, **J. Lee***, C. Park, H. Moon, H. Lim [arXiv] [Code] *arXiv, Sep 2026*

Where Activation Sparsity and KV-Cache Sparsity Cross in LLM Decoding
J. Lee, S. Lee, S. Hong, S. Eo, H. Lim [arXiv] [Code] *arXiv, Sep 2026*

Faster Block-Diffusion Serving with Distribution-Free Risk Guarantees
J. Lee, D.J. Lee, C. Park, S. Eo, H. Lim [arXiv] [Code] *arXiv, Sep 2026*

Vision Is Not Overhead: One-Pass Block Drafting for Lossless Speculative Decoding in Vision-Language Models
J. Lee, S. Hong, D.J. Lee, *et al.* [arXiv] [Code] *arXiv, Sep 2026*

Answer-Conditioned Chains of Thought Degrade Verifiable-Reasoning Distillation in Large Language Models
J. Lee, S. Lee, S. Son, *et al.* [arXiv] [Code] *arXiv, Jul 2026*

To Isolate or to Score? Model-Adaptive Assessment for Cost-Efficient Multi-Agent RAG
J. Lee, C. Park, H. Lim [arXiv] [Code] *arXiv, Jun 2026*

Beyond Penalizing Mistakes: Stabilizing Efficiency Training in Large Reasoning Models via Adaptive Correct-Only Rewards
J. Lee*, S. Lee*, S. Hong, *et al.* [arXiv] [Code] *arXiv, Jun 2026*

RenderRank: Learning to Rerank Text with Compressed Visual Tokens
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim [arXiv] *arXiv, Sep 2026*

LAMAR: An Open Language-Aware Multilingual Alignment Reranker
S. Hong, Y. Jang, **J. Lee**, S. Lee, H. Lim [arXiv] *arXiv, Jul 2026*

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